# How machines learn

AI · Level 1 · Lesson 2



# Today we will

- Put the five steps of machine learning in order.
- Explain why a model is tested on examples it has never seen.
- Explain why more, and more varied, examples help.

# Remember

- Collect examples, label them, train, test, then use the model.
- Test on examples kept aside: a perfect score on training examples can just mean memorising.
- A model only knows the kinds of examples it saw, so variety matters.

# Be the machine

1. Choose (3 min): Pick a secret pattern and keep it to yourself.
2. Train (6 min): Draw 8 shapes that follow it and 8 that don’t.
3. Test (6 min): Draw 6 new shapes; your partner predicts each one.
4. Score (4 min): Count the right answers, then reveal the pattern.
5. Swap and talk (6 min): Swap roles, then name the five steps you used.

**You need paper and a pen for each pair. Plan the secret pattern and the 6 test shapes before you start, and make sure at least one test shape looks different from all the training shapes.**

# Let’s discuss

1. What would you teach a machine to sort, and what labels would you use?
2. Why is memorising last year’s exam paper a risky way to revise?
3. Whose messages, voices or photos might be missing from examples collected only at your school?

# Exit ticket

- Write the five steps in order, and one reason to test on new examples.